Fiscal Year:	FY 2017	**Task Last Updated:**	FY 04/05/2017
PI Name:	German, Randall M. Ph.D.		
Project Title:	Multi-Scale Modeling and Experimentation on Liquid Phase Sintering in Gravity and Microgravity Environments (NNX16AK21G)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	co-investigator Eugene A Olevsky, SDSU, Elisa Torresani, SDSU, MS 1326		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2010 Materials Science NNH10ZTT001N
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No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Bhat, Biliyar	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	none		
COI Name (Institution):	Olevsky, Eugene Ph.D. (San Diego State University)		
Grant/Contract No.:	NNX16AK21G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	NOTE (Ed., June 2016): Continuation of "Multi-Scale Modeling and Experimentation on Liquid Phase Sintering in Gravity and Microgravity Environments", grant #NNX14AB31G with the same Principal Investigator. A research project is proposed to investigate the gravity-dependent densification, microstructure development, and the related macroscopic shape evolution of multi-phase powder materials during liquid phase sintering (LPS). The research is aimed at achieving two critical goals: (i) in-depth analysis of the LPS induced pore-grain structure evolution by the deconvolution of the gravity impact and (ii) exploring sintering under microgravity conditions as a technique for in-space fabrication and repair. Indeed, future NASA efforts to extend human exploration back to the Moon and beyond will require development of processes that permit fabrication under reduced gravity conditions. This capability is needed to reduce resource requirements for extraterrestrial construction and in space repair. Many proposed in-space fabrication and repair techniques involve particulate materials processing, such as via sintering, but limited data are available to		

	assess the viability of these proposed net-shape techniques under reduced gravity. The research team has developed a computer code for the multiple scale simulation of sintering, embracing both microstructure development and macroscopic densification and associated shape distortion. Liquid phase sintering experiments on Earth and in microgravity will be conducted for a variety of tungsten heavy alloys (W-Ni-Cu-Mn) to validate the sintering model framework. The anticipated research outcomes will be relevant to current and future space exploration needs for habitat creation, extraterrestrial exploration, and vehicle repair activities during trips to Mars or the Moon.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research is focused on liquid phase sintering, one of the most widely used manufacturing routes for ceramics, cemented carbides, powder metals, and composites. In the USA this is responsible for approximately $25 billion in annual product for components used in metal cutting, oil well drilling, automotive engines, motor bearings, computer heat sinks, jet engine valves, and even wrist watches (especially the newest cell phone coupled devices). These components are designed for high performance and high precision, and are often fabricated in large production quantities. The net-shape fabrication, where final size and shape is achieved without machining, is a challenge since often dimensions change 15 to 20% due to shrinkage in going from the powder to the dense component. During that high temperature sintering the component strength drops considerably and is subject to gravity effects. Thus, computer models of the sintering process are vital to proper tool and process design. Further, to attain specific mechanical, thermal, electrical, or other properties requires further attention to microstructure control. This research provides a verified computational model to enable prediction of final size and shape for different gravitational fields, using a multiple scale model with hopes to extend from geometry to property predictions. The robust inclusion of gravitational gradients, liquid-solid migration, and substrate friction, the computer simulation will be useful for predicting component attributes over a range of gravitational conditions, as required for eventual additive manufacturing as potentially envisioned for fabrication on the moon, other planets, or in microgravity, for example, in building structure on the lunar surface or repairing space vehicles. Of course, this has benefit on Earth with the evolution of a computational means to predict high tolerance shapes by sintering.
Task Progress:	Fundamentally this study is on how gravity acts to cause nonuniform response for particle systems that undergo high temperature heat treatment, known as sintering. The idea of sintering has been around for 26,000 years for the fabrication of bricks, pottery, and in the past 300 years, metals. Gravity cause a stress pointing downward on the object that naturally would cause "elephant foot" shapes. The object has friction on the bottom, so is pinned by friction but distorts. Today there are hundreds of composition only possible using sintering - tungsten carbide with cobalt, silicon nitride, titanium carbide, bronze bearings, and stainless steel filters. Our research is carefully extracting the gravity role during sintering. The research is showing how gravity plays a role in sintering densification (the pulling together of the particles by capillary force) including pore buoyancy and grain connectivity. Under reduced gravity, the sintered component dimensional accuracy decreases, densification decreases, and pores stabilize. At the same time, distortion increases since solid grains are less connected. Our research on predicting sintered dimensions relies on a matrix of liquid phase sintered tungsten alloys: seven times (0 to 60 min) at temperature for W-Cu-Ni-Mn compositions with density differences between solid (19.2 g/cc), liquid (about 10 g/cc), and pores (0 g/cc) to accentuate the role of gravity. The experiments focus is on different solid:liquid ratios and compositions which allow adjustment of the surface energies, as evident by microstructure dihedral angles. Microgravity experiments are taking place to isolate gravity's role in improving dimensional accuracy. The change in sintering trajectory with gravity is important at intermediate gravity levels, such as on the lunar surface or Mars. The findings are relevant to current industrial practice and anticipated space-based repair and fabrication concepts using additive manufacturing. The current work focuses on validation of new computational simulation models.
Bibliography Type:	Description: (Last Updated: 03/27/2025)
Articles in Peer-reviewed Journals	German RM. "The emergence of Quantitative Sintering Theory from 1945 to 1955." Journal of Metals (JOM). 2017 Apr;69(4):630-4. http://dx.doi.org/10.1007/s11837-016-2242-1 , Apr-2017
Articles in Peer-reviewed Journals	German RM. "Powder Metallurgy fundamentals: An introduction to the principles of sintering metal powders." Powder Metallurgy Review. 2016 Autumn/Fall;5 (3):71-80. http://www.pm-review.com/wp-content/uploads/2016/09/MAGAZINE-PMR-Autumn-2016-PDF-single-page-edition.pdf , Sep-2016
Papers from Meeting Proceedings	German RM, Johnson JL. "Replication Experiments in Microgravity Liquid Phase Sintering." Presented at PowderTech2016, Boston, Massachusetts, June 5-9, 2016. In: Advances in Powder Metallurgy and Particulate Materials 2016. Princeton, NJ : Metal Powder Industries Federation, 2016. p. 349-363. , Nov-2016